B - Material Engineering MS4092

* Wajib

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Number of problems: 8
Start at: 11:30 PM
Finish at: 11:48 PM
Duration: 18 Minutes

A new material X at 20 0C has an atomic radius 0.147 nm, an FCC crystal structure, and an atomic weight 25.751 g/mol. What is its theoretical density?

- ○ 1.511 g/cm^3
- ○ 8.81 g/cm^3
- ✕ 2.38 g/cm^3
- ○ 2.71 g/cm^3

FCC → 4 atom

Avogadro number = $6,022 \times 10^{23}$

$V = \frac{4}{3} \times \pi \times (0,147 \times 10^{-7})^3 = 1,33 \times 10^{-23}$

$m = \frac{25,751}{6,022 \times 10^{23}}\ g$ $\rho = \frac{4 \times \frac{25,751}{6.022 \times 10^{23}}}{4 \times 1,33 \times 10^{-23} \times 16,34}$

$\rho = 2.38\ g/cm^3$

Impact test curve of a metal represent...

- ○ Transition temperature ductile to brittle
- ○ Impact behaviour e.g. of Aluminium or steel vs temperature
- ✕ Transition temperature ductile to brittle and impact behaviour
- ○ Glass transition temperature

To verify that a metallic material such as steel, brass, aluminium alloy or other comply with an International standard of specification such as SAE J412 we can use...

- ✕ OES — optical emission spectroscopy
- ○ SEM and EDS — Energy Dispersive X-Ray Spectrometer & Scanning Electron Microscope
- ○ FTIR — Fourier transform infrared spectroscopy
- ○ XRD — X-Ray powder Diffraction

A spark plug cosist of alumina ceramic this is to...

- ○ Support the spark plug itself
- ○ Withstand a high temperature of the engine.
- ✕ Prevent the high voltage from the coil to jump to the engine body
- ○ Give a white colour

During the tensile practice at lab logam, the obtained modulus of elasticity value for a ST37 is usually lower than stated in the reference. This is caused by... *

- A slip plane is in the plane of 111
- A slip occurred between the extensometer and the sample
- A slip occurred between the sample and its grip.
- The sample is ST 41

Which statements regarding the elastic modulus is true?

elastic modulus ↑
semakin ductile
karbon ↑
makin brittle

- High carbon steel > Medium Carbon steel > Low carbon steel
- All answers are correct
- Steel with the higher carbon content has the higher of Elastic modulus
- Elastic modulus cannot be obtained from metallography

A cast iron is relatively easier and cheaper to manufacture for small industries than steel is due to...

- Lower melting temperature which is 1143 C
- Lower melting temperature which is 700 C
- Lower carbon content
- High melting temperature which is 1535 C

Example of ceramic material used in automotive is in the:

- Camshaft
- Piston in the combustion chamber
- Crankshaft
- Catalytic converter

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